

Iris Yang

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EDUCATION

Massachusetts Institute of Technology Cambridge, MA	2025-2026
GPA 4.9/5.0 <i>Master of Engineering in Electrical Engineering and Computer Science</i>	
Massachusetts Institute of Technology Cambridge, MA	2021-2025
GPA 4.6/5.0 <i>Dual Bachelor of Science in Computer Science and in Business Analytics</i>	

TECHNICAL SKILLS

Languages: Python, C, Java, JavaScript/TypeScript, R, HTML/CSS, SQL

Frontend: React, Vite, HTML/CSS, Figma

Backend & APIs: Node.js, Express, MongoDB, Python Flask, REST APIs, OAuth

ML / AI: PyTorch, TensorFlow, Scikit-Learn, HuggingFace Transformers, OpenCV, OpenAI API

Data & Analytics: Pandas, NumPy, Sawtooth, Matplotlib

PROJECTS

Master's Thesis - Causal Framework for Attribution in Vision Transformers <i>PyTorch</i>	September 2025–May 2026
- Developed a multi-granularity causal framework for cross-validating ViT interpretability methods - Found propagation-based attribution to be more causally grounded than gradient-based methods - Demonstrated no tradeoff between causal accuracy and perceptual interpretability in ViT attribution	
PlantIt - Full Stack Team Task Scheduler <i>TS, HTML/CSS, Vite, MongoDB, Node.js, OAuth</i>	December 2024
- Designed a task-dependency engine with cycle detection, task prioritization, and completion tracking across linked tasks - Built and shipped the full stack end-to-end, from user research and architecture design through implementation and testing - Modeled many-to-many user-task relationships with a table/timeline view for team-wide scheduling and visibility - Link: https://plant-it-beta.vercel.app/	
ThreadChef - Agentic LLM Cooking Assistant <i>Llama 3.1 8B, Gradio, HuggingFace Spaces</i>	May 2026
- Built and deployed an LLM scheduling assistant that converts text-based recipes into verified, parallelized cooking schedules - Implemented internal Python checks for LLM output to enforce graph-consistency, with corrective re-prompting on failure - Validated extraction against hand-labeled benchmark at 91% classification accuracy - Link: https://huggingface.co/spaces/ixyang/threadchef	
Dynamic Beam Width for LLM Decoding <i>GPT-2, PyTorch</i>	December 2024
- Developed a self-adjusting beam search algorithm driven by token-level attention entropy and perplexity - Achieved up to 1.9x faster decoding speed vs. standard beam search while maintaining comparable output quality - Link: https://irisxyang.github.io/beam_search_paper.pdf	
PixelThis - Full Stack Web App <i>JS, HTML/CSS, React, MongoDB, Node.js, Figma, OAuth</i>	January 2023
- Built a full-stack social media web app from scratch with OAuth-authenticated users and posting functionality - Engineered an interactive canvas rendering feature and state handling to support drawing and saving creations - Link: https://pixel-thls.herokuapp.com/	
Handwriting Transcription App <i>Google CV API, OpenCV, Python, Flask</i>	April 2024
- Built end-to-end OCR pipeline with Google Cloud Vision API + OpenCV, served through a Python Flask UI - Achieved 90% accuracy on transcription and 95% sketch extraction through adaptive thresholding and contour detection in OpenCV - Link: https://github.com/irisxyang/htr-proj	
Morgan Stanley 2022 Code To Give Hackathon <i>JS, HTML/CSS, Figma</i>	March 2022
- Won 2nd place of 100+ participants - Designed and prototyped a front-end website interface to aid those facing homelessness	

EXPERIENCE

Graduate Teaching Assistant - MIT 6.104 Software Design Cambridge, MA	June 2025–January 2026
- Taught full-stack software design and implementation to 100+ students, covering LLM-API integration across frontend and backend - Led detailed feedback sessions for AI-integrated web apps, focusing on code structure, error handling, and RESTful API design	
Data Science Researcher - MIT CISR Cambridge, MA	May 2024–January 2025
- Helped scope the research questions and data requirements for a study of data-monetization practices across 300+ global firms - Built a reusable Python pipeline to clean, analyze, and benchmark data for evaluating firm performance	
Machine Learning Researcher - MIT CSAIL Remote	May 2023–July 2023
- Implemented data processing modules in Python to enable a semi-automated ML extraction pipeline - Designed 20+ templates in Python for parsing and extracting electronic product specs with 98% accuracy	

Economics and Data Science Researcher – MIT Media Lab | Cambridge, MA

May 2022–September 2022

- Identified optimal store-closure scenarios projected to improve maximum profit by 30%
- Designed data filtering pipeline in Python to calibrate an enhanced Huff model for predicting customer foot traffic

INVOLVEMENT AND INTERESTS

Activities: MIT Varsity Swimming, Society of Women Engineers at MIT, MIT 2025 Ring Committee

Languages: English and Chinese (Fluent), French (Intermediate)

Interests: Visual arts (Portfolio: <http://irisxyang.github.io/portfolio>), cycling, lifting, coffee